

Physical Chemistry–III (CHB-504)

Unit 1: Thermodynamics of Mixtures

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It's hot but studying is also important

Contents

1	Partial Molar Quantities	3
1.1	Definition of Partial Molar Volume	3
1.2	Why Partial Molar Volume Can Be Negative	3
1.3	Water–Ethanol System: Variation of $\bar{V}$ with Composition	3
1.4	Partial Molar Gibbs Energy (Chemical Potential)	3
2	Chemical Potential (μ_i)	4
2.1	Physical Significance of Chemical Potential	4
2.2	Chemical Potential of Ideal Gas	4
2.3	Variation of Chemical Potential with Temperature and Pressure	4
2.4	Determining Chemical Potential from a G vs. n Plot	5
3	Gibbs–Duhem Equation	5
3.1	Derivation	5
3.2	At Constant Temperature and Pressure	5
3.3	Significance of the Gibbs–Duhem Equation	6
4	Fugacity	6
4.1	Physical Significance and Definition	6
4.2	Derivation of Relation among Fugacity, Temperature, and Pressure	7
4.3	Relation between Fugacity and the Virial Equation	7
4.4	Worked Example: Fugacity of N_2 at 27°C , 100 atm	7
5	Lewis–Randall Rule	8
5.1	Statement	8
5.2	Derivation	8
5.3	Significance and Applicability	8
6	Gibbs–Duhem–Margules Equation	8
6.1	Derivation	8
6.2	Application: Connection between Raoult's and Henry's Laws	9

7	Thermodynamics of Mixing of Ideal Gases	9
7.1	Setup	9
7.2	Gibbs Energy of Mixing	9
7.3	Entropy of Mixing	10
7.4	Enthalpy of Mixing	10
7.5	Volume of Mixing	10
7.6	Worked Example	10
8	Free Energy of Mixing and Spontaneity	10
9	Duhem–Margules Equation and Its Applications	11
9.1	Statement	11
9.2	Proof That Raoult’s Law Applied to One Component Implies Henry’s Law for the Other	11
9.3	Relation to Raoult’s and Henry’s Laws	11
10	Fundamental Equation of Chemical Thermodynamics	11
10.1	Derivation	11
10.2	Non-Expansion Work from Changing Composition	12
10.3	Chemical Potential from Other Thermodynamic Potentials	12
11	Summary of Key Results	12

Partial Molar Quantities

Definition of Partial Molar Volume

Definition: The partial molar volume $\bar{V}_i$ of component i in a mixture is defined as the rate of change of total volume V with respect to the amount (moles) of component i , at constant temperature, pressure, and amounts of all other components:

$$\bar{V}_i = \left(\frac{\partial V}{\partial n_i} \right)_{T, P, n_{j \neq i}}$$

Physically, $\bar{V}_i$ represents the contribution of one mole of species i to the *total* volume of the mixture, at the given composition. It is **not** the same as the molar volume of pure i because molecular interactions between unlike species change the effective volume.

Why Partial Molar Volume Can Be Negative

The total volume of a mixture is:

$$V = n_A \bar{V}_A + n_B \bar{V}_B$$

Molar volumes of pure substances are always positive. However, in a mixture, strong solute–solvent interactions can cause the solution to occupy *less* space than the pure components. Therefore $\bar{V}_i$ can be less than V_i^* (pure molar volume) and in extreme cases even negative (e.g., MgSO_4 in water at dilute concentrations).

Water–Ethanol System: Variation of $\bar{V}$ with Composition

When water and ethanol are mixed:

- At pure water end: $\bar{V}_{\text{water}} \approx 18 \text{ cm}^3 \text{ mol}^{-1}$ (pure molar volume).
- As ethanol is added, $\bar{V}_{\text{water}}$ *decreases* below $18 \text{ cm}^3 \text{ mol}^{-1}$ because ethanol molecules fit into the open hydrogen-bonded network of water, reducing the total volume.
- Similarly, $\bar{V}_{\text{ethanol}}$ changes with composition.

The classic statement:

“Partial molar volume of water in water–ethanol mixtures is $18 \text{ cm}^3 \text{ mol}^{-1}$ in pure water but falls to $\approx 14 \text{ cm}^3 \text{ mol}^{-1}$ in ethanol-rich mixtures.”

This is because in the ethanol-rich environment, water molecules are surrounded by ethanol, which causes significant contraction due to hydrogen bonding.

Partial Molar Gibbs Energy (Chemical Potential)

Among all partial molar quantities, the most important is the partial molar Gibbs energy, called the **chemical potential**:

$$\mu_i = \bar{G}_i = \left(\frac{\partial G}{\partial n_i} \right)_{T, P, n_{j \neq i}}$$

The total Gibbs energy is:

$$G = \sum_i n_i \mu_i$$

Chemical Potential (μ_i)

Physical Significance of Chemical Potential

The chemical potential μ_i of component i in a system is the partial molar Gibbs energy. Its physical significance is multi-fold:

1. **Criterion of Phase Equilibrium:** At equilibrium between two phases α and β :

$$\mu_i^{(\alpha)} = \mu_i^{(\beta)}$$

A species migrates from a phase of higher μ_i to lower μ_i until equilibrium is established.

2. **Driving Force for Mass Transfer:** Like temperature drives heat flow and pressure drives mechanical expansion, the chemical potential drives mass (particle) transfer between phases or regions.

3. **Criterion of Chemical Equilibrium:** For a reaction at equilibrium:

$$\sum_i \nu_i \mu_i = 0$$

where ν_i are stoichiometric coefficients (positive for products, negative for reactants).

4. **Relation to Escaping Tendency:** A higher μ_i means a greater tendency for the species to escape from the phase (evaporate, dissolve, react).

Chemical Potential of Ideal Gas

For a pure ideal gas:

$$\mu = \mu^\circ + RT \ln \left(\frac{P}{P^\circ} \right)$$

For component i in an ideal gas mixture (Dalton's law: partial pressure $p_i = x_i P$):

$$\mu_i = \mu_i^\circ + RT \ln \left(\frac{p_i}{P^\circ} \right)$$

Variation of Chemical Potential with Temperature and Pressure

From the fundamental equation $G = H - TS$ and $dG = -S dT + V dP + \sum \mu_i dn_i$:

With Temperature (at constant P):

$$\left(\frac{\partial \mu_i}{\partial T} \right)_{P,n} = -\bar{S}_i$$

Since entropy $\bar{S}_i > 0$, chemical potential *decreases* with increasing temperature.

With Pressure (at constant T):

$$\left(\frac{\partial \mu_i}{\partial P} \right)_{T,n} = \bar{V}_i$$

Since $\bar{V}_i > 0$, chemical potential *increases* with increasing pressure.

Determining Chemical Potential from a G vs. n Plot

From the plot of total Gibbs energy G vs. amount n of a substance at constant T , P , and amounts of other components:

$$\mu = \left(\frac{\partial G}{\partial n} \right)_{T,P,n_j} = \text{slope of the tangent to the } G\text{--}n \text{ curve}$$

The chemical potential equals the instantaneous slope (tangent) of the G vs. n curve at the composition of interest.

Gibbs–Duhem Equation

Derivation

Starting from the fundamental equation for the Gibbs energy of a multi-component system. At constant T and P , the Gibbs energy depends on composition:

$$G = G(T, P, n_1, n_2, \dots)$$

By the Euler theorem for extensive functions (since G is extensive and μ_i are intensive):

$$G = \sum_i n_i \mu_i \quad (1)$$

Taking the total differential of equation (1):

$$dG = \sum_i n_i d\mu_i + \sum_i \mu_i dn_i$$

But from thermodynamics, the fundamental equation at variable composition is:

$$dG = -S dT + V dP + \sum_i \mu_i dn_i$$

Comparing both expressions for dG :

$$-S dT + V dP + \sum_i \mu_i dn_i = \sum_i n_i d\mu_i + \sum_i \mu_i dn_i$$

Cancelling $\sum_i \mu_i dn_i$ from both sides:

$$\boxed{-S dT + V dP - \sum_i n_i d\mu_i = 0}$$

This is the **Gibbs–Duhem equation** in its general form.

At Constant Temperature and Pressure

At constant T and P ($dT = 0$, $dP = 0$):

$$\boxed{\sum_i n_i d\mu_i = 0}$$

Or equivalently, in terms of mole fractions $x_i = n_i/n$:

$$\sum_i x_i d\mu_i = 0$$

For a binary mixture:

$$x_A d\mu_A + x_B d\mu_B = 0 \implies d\mu_A = -\frac{x_B}{x_A} d\mu_B$$

Significance of the Gibbs–Duhem Equation

1. **Constraint on μ_i :** The chemical potentials of components in a mixture are *not independent*. If the chemical potential of one component changes, those of the others must adjust according to the Gibbs–Duhem equation.
2. **Thermodynamic Consistency Test:** Used to check whether experimentally determined activity coefficients are self-consistent (thermodynamic consistency test of Redlich–Kister).
3. **Derivation of Duhem–Margules Equation:** Leads to the Duhem–Margules equation relating the vapour pressures of the two components in a binary mixture:

$$x_A \left(\frac{\partial \ln p_A}{\partial x_A} \right)_{T,P} + x_B \left(\frac{\partial \ln p_B}{\partial x_A} \right)_{T,P} = 0$$

4. **Relation between Raoult’s and Henry’s Laws:** If one component of a binary mixture obeys Raoult’s law (at all compositions), the Gibbs–Duhem equation requires the other to obey Henry’s law in the dilute limit.

Fugacity

Physical Significance and Definition

For a real gas, the chemical potential cannot be expressed simply in terms of pressure because the gas does not behave ideally. G.N. Lewis introduced the concept of **fugacity** f as an *effective pressure* that corrects for non-ideal behaviour:

$$\mu = \mu^\circ + RT \ln \left(\frac{f}{f^\circ} \right)$$

where f° is the fugacity in the standard state. The fugacity has units of pressure. It measures the *escaping tendency* of a real gas.

Limiting condition:

$$\lim_{P \rightarrow 0} \frac{f}{P} = 1 \quad (\text{real gas} \rightarrow \text{ideal gas at low pressure})$$

Derivation of Relation among Fugacity, Temperature, and Pressure

For a real gas at constant T :

$$d\mu = \bar{V} dP = \frac{RT}{P} dP \quad (\text{ideal})$$

For a real gas, defining fugacity through:

$$d\mu = RT d \ln f \quad \text{at constant } T$$

Integrating from a low pressure P^* (where the gas is ideal, $f^* \approx P^*$) to pressure P :

$$\mu(T, P) - \mu(T, P^*) = RT \ln \frac{f}{P^*}$$

Also from exact thermodynamics:

$$\mu(T, P) - \mu(T, P^*) = \int_{P^*}^P \bar{V} dP$$

Therefore:

$$RT \ln \frac{f}{P^*} = \int_{P^*}^P \bar{V} dP$$

Taking $P^* \rightarrow 0$:

$$RT \ln f = \int_0^P \bar{V} dP + RT \ln P^* - \int_0^{P^*} \frac{RT}{P} dP$$

A cleaner standard form using the compressibility factor $Z = P\bar{V}/RT$:

$$RT \ln \frac{f}{P} = \int_0^P \frac{(Z - 1)}{P} RT dP$$

$$\boxed{\ln \frac{f}{P} = \frac{1}{RT} \int_0^P \left(\bar{V} - \frac{RT}{P} \right) dP}$$

Relation between Fugacity and the Virial Equation

Using the virial equation of state truncated at the second virial coefficient B :

$$\bar{V} = \frac{RT}{P} + B$$

Substituting:

$$\ln \frac{f}{P} = \frac{1}{RT} \int_0^P B dP = \frac{BP}{RT}$$

$$\boxed{f = P \exp\left(\frac{BP}{RT}\right)}$$

Worked Example: Fugacity of N_2 at 27°C, 100 atm

Given: $B = -0.016 \text{ L mol}^{-1}$, $T = 300 \text{ K}$, $P = 100 \text{ atm}$, $R = 0.0821 \text{ L atm K}^{-1} \text{ mol}^{-1}$.

$$\frac{BP}{RT} = \frac{(-0.016)(100)}{(0.0821)(300)} = \frac{-1.6}{24.63} = -0.06496$$

$$f = 100 \times e^{-0.06496} = 100 \times 0.9371 \approx \boxed{93.7 \text{ atm}}$$

Lewis–Randall Rule

Statement

The **Lewis–Randall rule** states that the fugacity f_i of component i in an ideal mixture is equal to the product of its mole fraction x_i and the fugacity f_i^* of the pure component at the same temperature and pressure:

$$f_i = x_i f_i^*$$

Derivation

An ideal mixture (or ideal solution in the gas phase) is defined such that each component i obeys:

$$\mu_i(T, P, x) = \mu_i^*(T, P) + RT \ln x_i$$

where $\mu_i^*(T, P)$ is the chemical potential of pure i at the same T and P .

Using $\mu_i = \mu_i^\circ + RT \ln f_i$ and $\mu_i^* = \mu_i^\circ + RT \ln f_i^*$:

$$RT \ln f_i = RT \ln f_i^* + RT \ln x_i$$

$$\therefore f_i = x_i f_i^*$$

Significance and Applicability

- For **ideal gas mixtures**: since $f_i^* = P$, the rule gives $f_i = x_i P = p_i$ (Dalton's law).
- The rule is a good approximation for mixtures of gases with similar molecular properties (e.g., $\text{N}_2 + \text{O}_2$).
- For real mixtures, it fails because unlike intermolecular interactions are not accounted for.

Gibbs–Duhem–Margules Equation

Derivation

Starting from the Gibbs–Duhem equation at constant T and P for a binary system:

$$x_A d\mu_A + x_B d\mu_B = 0$$

Using $\mu_i = \mu_i^\circ + RT \ln f_i$:

$$d\mu_i = RT d \ln f_i$$

Substituting:

$$x_A d \ln f_A + x_B d \ln f_B = 0$$

For variations with composition at constant T and P (differentiating with respect to $\ln x_A$):

$$x_A \left(\frac{\partial \ln f_A}{\partial \ln x_A} \right)_{P,T} + x_B \left(\frac{\partial \ln f_B}{\partial \ln x_B} \right)_{P,T} = 0$$

Since $x_A + x_B = 1 \Rightarrow dx_A = -dx_B$, dividing by dx_A :

$$x_A \left(\frac{\partial \ln f_A}{\partial x_A} \right)_{P,T} = x_B \left(\frac{\partial \ln f_B}{\partial x_B} \right)_{P,T}$$

Application: Connection between Raoult's and Henry's Laws

Replacing f_i by partial pressure p_i (valid for low pressures):

$$x_A \left(\frac{\partial \ln p_A}{\partial x_A} \right)_T = x_B \left(\frac{\partial \ln p_B}{\partial x_B} \right)_T$$

If component A obeys Raoult's law: $p_A = x_A p_A^*$, then $\ln p_A = \ln x_A + \ln p_A^*$, so:

$$\left(\frac{\partial \ln p_A}{\partial \ln x_A} \right)_T = 1 \implies x_A \left(\frac{\partial \ln p_A}{\partial x_A} \right)_T = 1$$

Therefore for component B:

$$x_B \left(\frac{\partial \ln p_B}{\partial x_B} \right)_T = 1 \implies p_B = K_H x_B \quad (\text{Henry's law})$$

Conclusion: If one component obeys Raoult's law, the other must obey Henry's law in the dilute limit.

Thermodynamics of Mixing of Ideal Gases**Setup**

Consider n_A moles of ideal gas A and n_B moles of ideal gas B, initially separated, at the same temperature T and pressure P . They are allowed to mix at constant T and P .

Chemical potential of i before mixing (pure gas at P):

$$\mu_i^{\text{pure}} = \mu_i^\circ + RT \ln P$$

Chemical potential of i after mixing (partial pressure $p_i = x_i P$):

$$\mu_i^{\text{mix}} = \mu_i^\circ + RT \ln(x_i P) = \mu_i^\circ + RT \ln P + RT \ln x_i$$

Gibbs Energy of Mixing

$$\begin{aligned} \Delta_{\text{mix}} G &= G_{\text{final}} - G_{\text{initial}} = \sum_i n_i (\mu_i^{\text{mix}} - \mu_i^{\text{pure}}) \\ &= \sum_i n_i RT \ln x_i \end{aligned}$$

$$\boxed{\Delta_{\text{mix}} G = RT \sum_i n_i \ln x_i = nRT \sum_i x_i \ln x_i}$$

Since $0 < x_i < 1$, we have $\ln x_i < 0$, therefore:

$$\Delta_{\text{mix}} G < 0$$

Mixing is spontaneous at constant T and P (consistent with entropy considerations).

Entropy of Mixing

$$\Delta_{\text{mix}}S = - \left(\frac{\partial \Delta_{\text{mix}}G}{\partial T} \right)_{P, n_i} = -R \sum_i n_i \ln x_i$$

$$\Delta_{\text{mix}}S = -nR \sum_i x_i \ln x_i > 0$$

This is always **positive** (mixing increases disorder), which is why mixing is spontaneous.

Enthalpy of Mixing

$$\Delta_{\text{mix}}H = \Delta_{\text{mix}}G + T \Delta_{\text{mix}}S = nRT \sum_i x_i \ln x_i + T \left(-nR \sum_i x_i \ln x_i \right)$$

$$\Delta_{\text{mix}}H = 0$$

For an ideal gas mixture, there is **no enthalpy change on mixing** because there are no intermolecular interactions. Mixing is driven entirely by entropy.

Volume of Mixing

$$\Delta_{\text{mix}}V = \left(\frac{\partial \Delta_{\text{mix}}G}{\partial P} \right)_{T, n_i} = 0$$

$$\Delta_{\text{mix}}V = 0$$

No change in volume on mixing ideal gases.

Worked Example

Calculate $\Delta_{\text{mix}}G$ and $\Delta_{\text{mix}}S$ when 1 mol hexane (C_6H_{14}) is mixed with 1 mol heptane (C_7H_{16}) at 298 K (treat as ideal solution).

$$n = 2 \text{ mol}, x_A = x_B = 0.5$$

$$\begin{aligned} \Delta_{\text{mix}}G &= (2)(8.314)(298)(0.5 \ln 0.5 + 0.5 \ln 0.5) = 2 \times 8.314 \times 298 \times \ln 0.5 \\ &= 2 \times 8.314 \times 298 \times (-0.6931) = -3434 \text{ J mol}^{-1} \approx -3.43 \text{ kJ} \end{aligned}$$

$$\Delta_{\text{mix}}S = - \left(\frac{\partial \Delta_{\text{mix}}G}{\partial T} \right)_P = -nR \sum_i x_i \ln x_i = -2 \times 8.314 \times (-0.6931) = +11.53 \text{ J K}^{-1}$$

$$\Delta_{\text{mix}}H = 0$$

Free Energy of Mixing and Spontaneity

The three conditions for spontaneous isothermal–isobaric mixing are:

$$\Delta_{\text{mix}}G < 0 \quad \checkmark$$

$$\Delta_{\text{mix}}S > 0 \quad \checkmark$$

$$\Delta_{\text{mix}}H = 0 \quad (\text{ideal}; \Delta H < 0 \text{ also makes mixing more spontaneous})$$

Graphically, the $\Delta_{\text{mix}}G$ vs. mole fraction curve is a *concave downward* parabola with minimum at $x = 0.5$ (for a symmetric binary mixture). Any composition on this curve has lower Gibbs energy than the unmixed state, confirming spontaneous mixing.

Duhem–Margules Equation and Its Applications

Statement

For a binary liquid mixture at constant T and P :

$$x_A \left(\frac{\partial \ln p_A}{\partial x_A} \right)_{T,P} + x_B \left(\frac{\partial \ln p_B}{\partial x_A} \right)_{T,P} = 0$$

Equivalently:

$$x_A d \ln p_A + x_B d \ln p_B = 0$$

Proof That Raoult's Law Applied to One Component Implies Henry's Law for the Other

(See Section 6.2 above for the full derivation via the Gibbs–Duhem–Margules equation.)

Relation to Raoult's and Henry's Laws

Raoult's Law (solvent in dilute solution):

$$p_A = x_A p_A^* \quad (x_A \rightarrow 1)$$

Henry's Law (solute in dilute solution):

$$p_B = K_H x_B \quad (x_B \rightarrow 0)$$

These two are not independent – the Gibbs–Duhem equation shows they must coexist in the same ideal-dilute solution.

Fundamental Equation of Chemical Thermodynamics

Derivation

The Gibbs energy of a closed system is $G = H - TS$, with $dH = T dS + V dP$ for a reversible process. Thus:

$$dG = dH - T dS - S dT = V dP - S dT$$

For an **open system** where composition can change:

$$dG = -S dT + V dP + \sum_i \mu_i dn_i$$

This is the **fundamental equation of chemical thermodynamics**.

Non-Expansion Work from Changing Composition

At constant T and P (so $dT = dP = 0$):

$$dG = \sum_i \mu_i dn_i$$

For a spontaneous chemical change, $dG < 0$ means the system can do work. The **maximum non-expansion work** obtainable equals:

$$w_{\text{add, max}} = \Delta G = \sum_i \mu_i \Delta n_i$$

This is the basis of electrochemical cells: chemical composition change drives electrical work.

Chemical Potential from Other Thermodynamic Potentials

$$\mu_i = \left(\frac{\partial G}{\partial n_i} \right)_{T,P,n_j} = \left(\frac{\partial A}{\partial n_i} \right)_{T,V,n_j} = \left(\frac{\partial U}{\partial n_i} \right)_{S,V,n_j} = \left(\frac{\partial H}{\partial n_i} \right)_{S,P,n_j}$$

Summary of Key Results

Quantity	Expression
Chemical potential	$\mu_i = \left(\frac{\partial G}{\partial n_i} \right)_{T,P,n_j}$
Ideal gas μ_i	$\mu_i = \mu_i^\circ + RT \ln(p_i/P^\circ)$
Gibbs–Duhem (const. T, P)	$\sum_i n_i d\mu_i = 0$
Fugacity relation	$\mu = \mu^\circ + RT \ln f$
Lewis–Randall rule	$f_i = x_i f_i^*$
Fugacity (virial)	$f = P \exp(BP/RT)$
$\Delta_{\text{mix}} G$ (ideal)	$nRT \sum_i x_i \ln x_i < 0$
$\Delta_{\text{mix}} S$ (ideal)	$-nR \sum_i x_i \ln x_i > 0$
$\Delta_{\text{mix}} H$ (ideal)	0
$\Delta_{\text{mix}} V$ (ideal)	0

Exam Tip: Every derivation in this unit flows from two starting points: (1) the fundamental equation $dG = -SdT + VdP + \sum \mu_i dn_i$, and (2) the ideal gas expression $\mu_i = \mu_i^\circ + RT \ln p_i$. Master these two and all other results follow.